

Structural Obfuscation and Crypto-Steganography-Based Secured JPEG Compression Hardware for Medical Imaging Systems

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Introduction

- The electronic integrated circuits (ICs) and internet technology have eased and fastened the diagnosis and treatment of critical diseases.
- The medical instruments such as computed Tomography (CT) scanner and magnetic resonance imaging (MRI) scanner generate large size of digital image data [1].
- Due to large, these images are required to be stored/ transmitted in the compressed form.

[1] D. A. Koff and H. Shulman, "An overview of digital compression of medical images: Can we use lossy image compression in radiology?" Can. Assoc. Radiol. J., vol. 57, no. 4, pp. 211217, 2006.

Threat Model

- Emerging hardware threats are increasingly corrupting compressed medical images from cameras and scanners, often by compromising the compression processor.
- Hardware threat like reverse engineering (causing Trojan insertion) [7] and piracy [8]-[10].



[7] X. Zhang and M. Tehranipoor, "Case study: Detecting hardware trojans in third-party digital IP cores," in Proc. IEEE Int. Symp. Hardw.-Oriented Secur. Trust, San Diego CA, USA, Jun. 2011, pp. 6770.

[8] Maxim. Accessed: May. [Online]. Available: <https://www.maximintegrated.com/en/app-notes/index.mvp/id/545>

[9] SMT Corp. Counterfeit Detection. Accessed: May 2019. [Online]. Available: <https://www.smtcorp.com/counterfeit-detection>

[10] A. Sengupta, D. Roy, and S. P. Mohanty, "Triplephase watermarking for reusable ip core protection during architecture synthesis," IEEE Trans. Comput.-Aided Design Integr. Circuits Syst., vol. 37, no. 4, pp. 742755, Apr. 2018..

Related Work

Sr. No.	Existing Work	Technique Used	Remark
1.	F. Koushanfar et.al., [17] (2005) B. Le Gal et.al.,[22](2012)	employed watermarking technique to secure the designs against counterfeiting/cloning.	However, [17],[22] did not ensure the preventive security against RE
2.	A. Sengupta et.al., [11] (2017) A. Sengupta et.al., [20] (2005)	structural obfuscation has been applied on DCT	However, [11], [20] did not perform structural obfuscation on entire JPEG CODEC processor

- [17] F. Koushanfar, I. Hong, and M. Potkonjak, "Behavioral synthesis techniques for intellectual property protection," ACM Trans. Des. Autom. Electron. Syst., vol. 10, no. 3, pp. 523545, Jul. 2005.
- [22] B. Le Gal and L. Bossuet, "Automatic low-cost IP watermarking technique based on output mark insertions," Des. Autom. Embedded Syst., vol. 16, no. 2, pp. 7192, Jun. 2012.
- [20] A. Sengupta and M. Rathor, "Protecting DSP kernels using robust hologrambased obfuscation," IEEE Trans. Consum. Electron., vol. 65, no. 1, pp. 99108, Feb. 2019.
- [11] A. Sengupta, D. Roy, S. P. Mohanty, and P. Corcoran, "DSP design protection in CE through algorithmic transformation based structural obfuscation," IEEE Trans. Consum. Electron., vol. 63, no. 4, pp. 467476, Nov. 2017.

Proposed Methodology: Overview

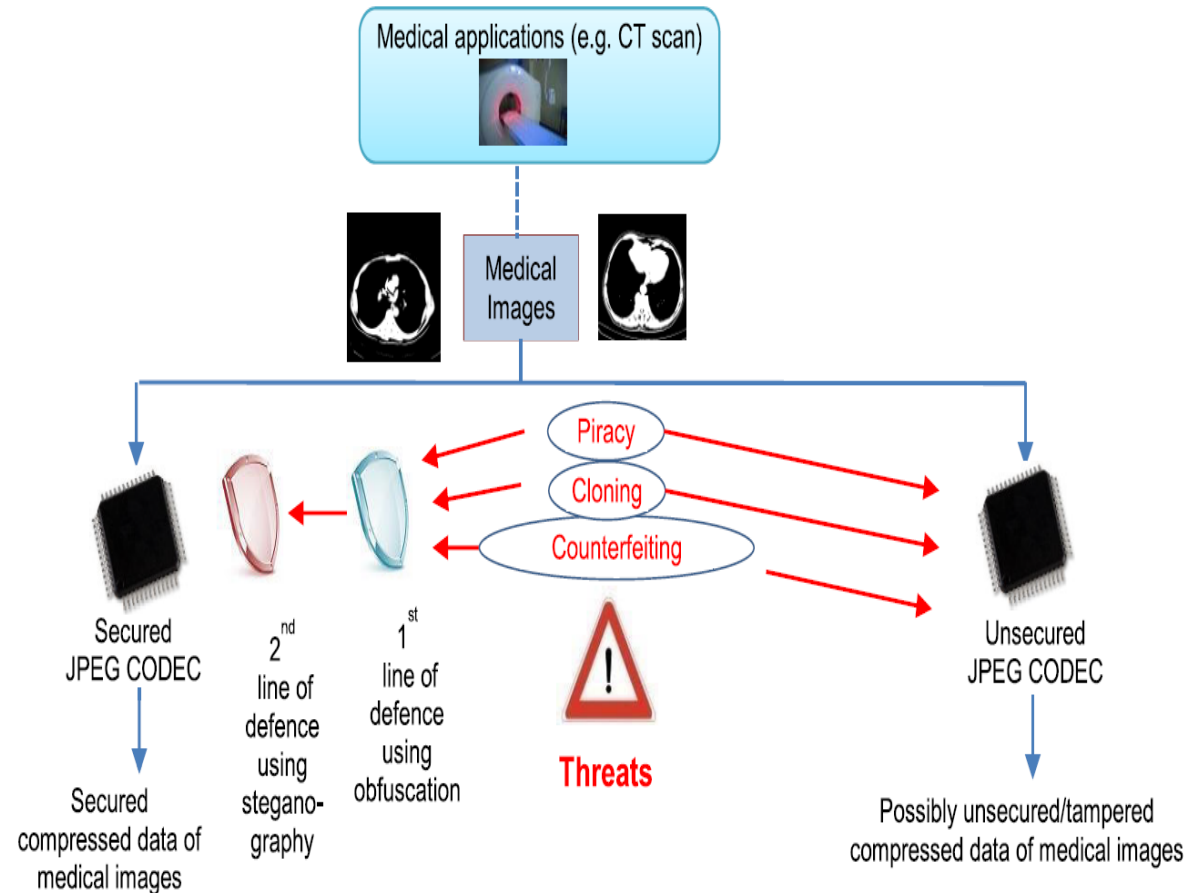


FIGURE 1. Thematic representation of securing medical images against external hardware threats.

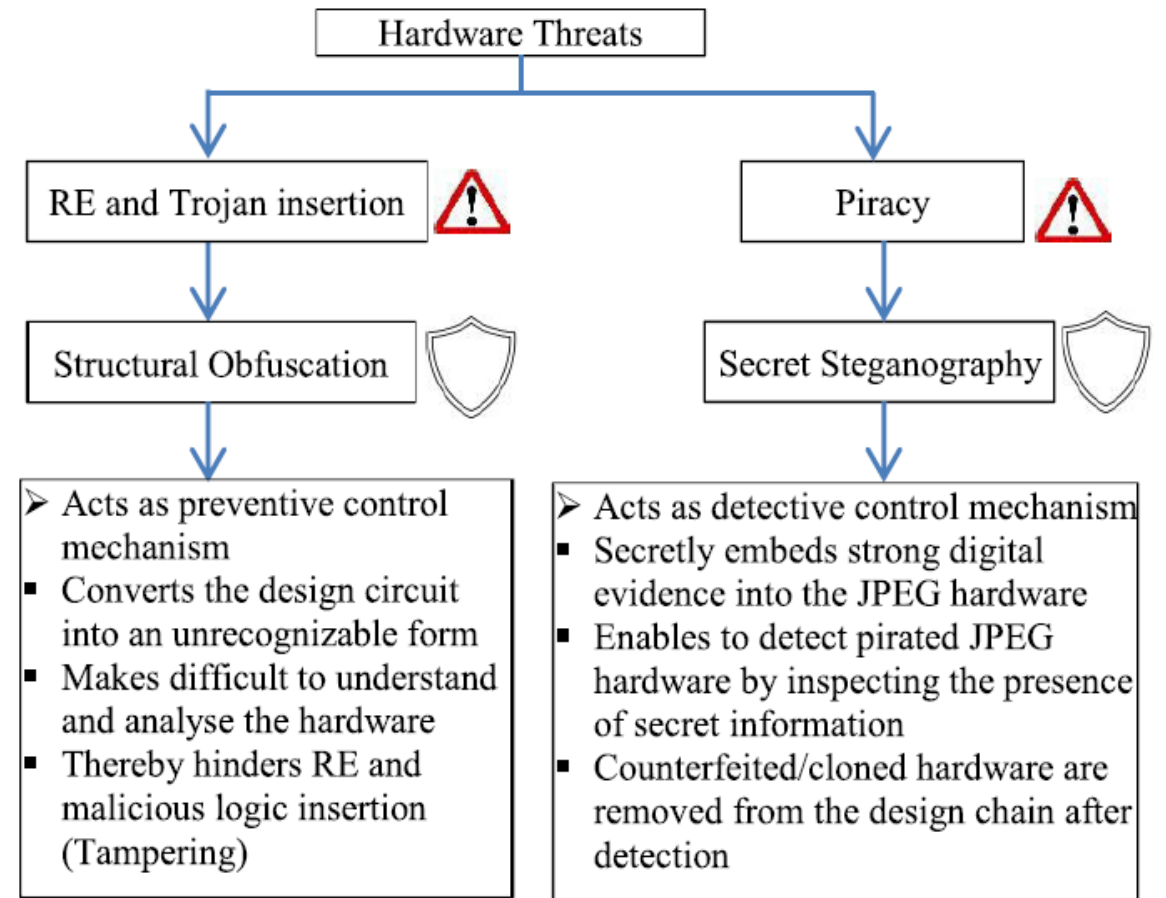


FIGURE 2.. Overview of the hardware threat and protection scenarios using proposed approach.

Details of the Proposed Approach

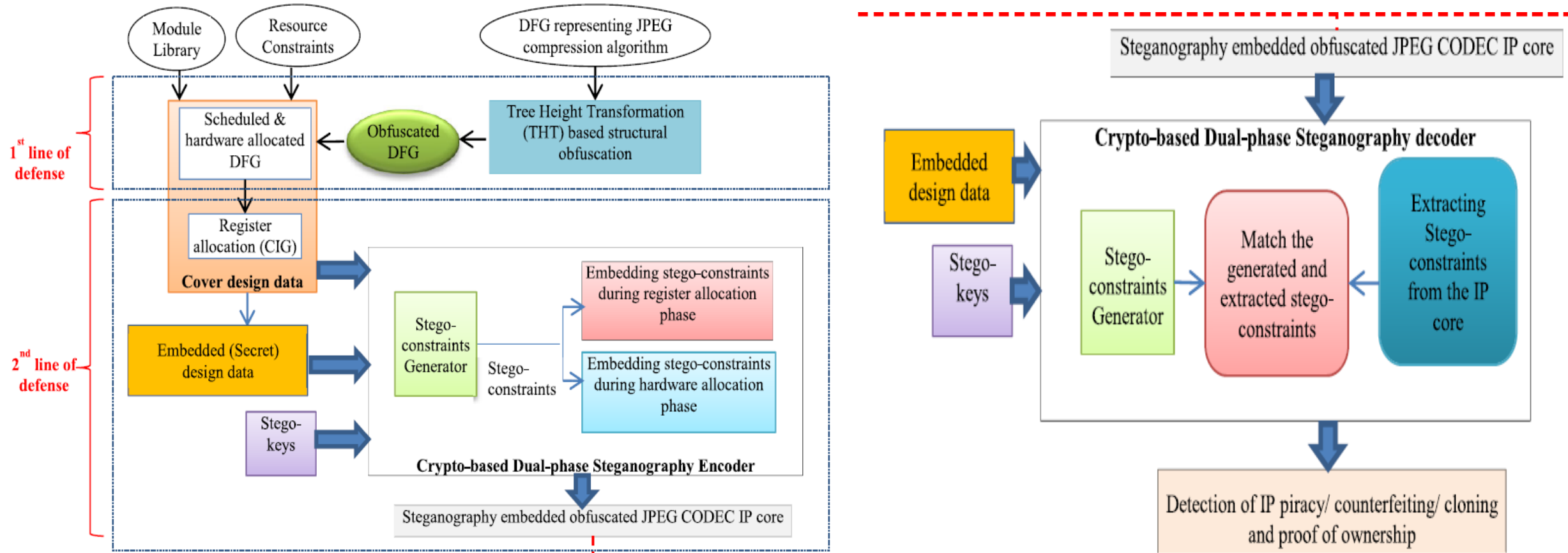


FIGURE 3(a). Proposed double line defense using structural obfuscation and crypto-based dual-phase steganography to secure JPEG CODEC processor used for compression of medical images.

Details of the Proposed Approach

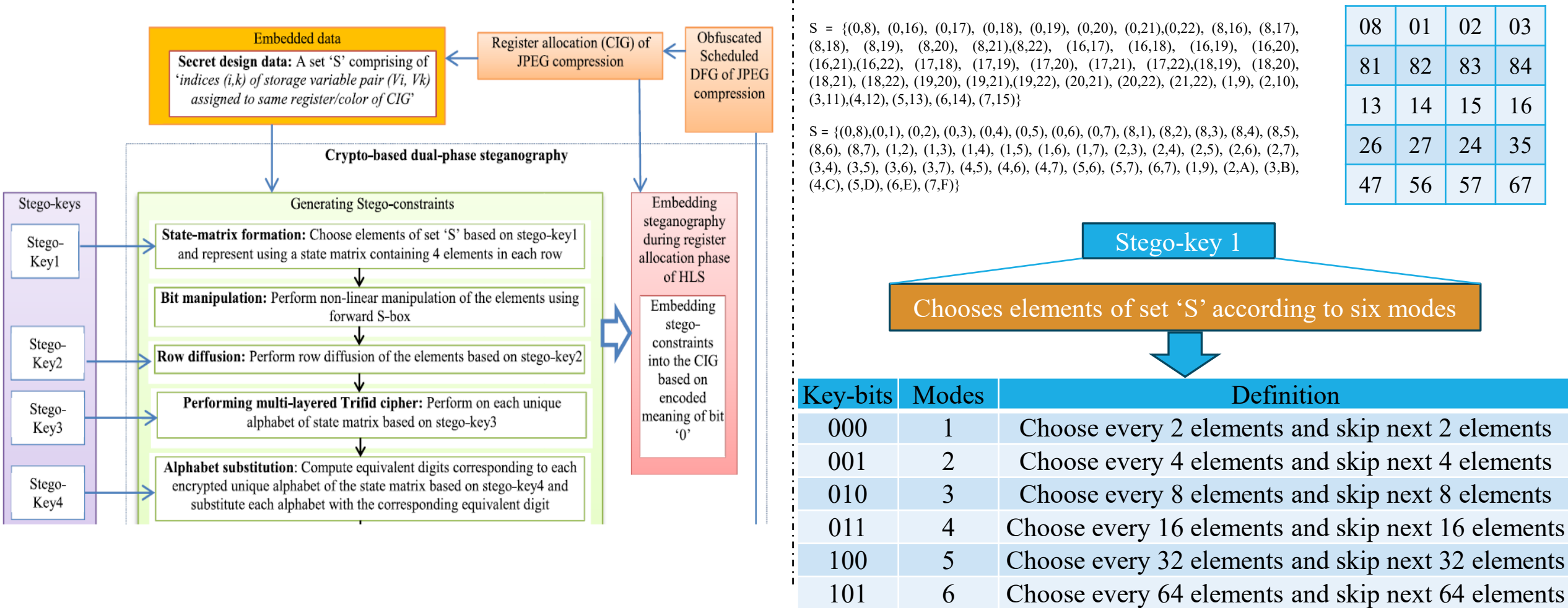
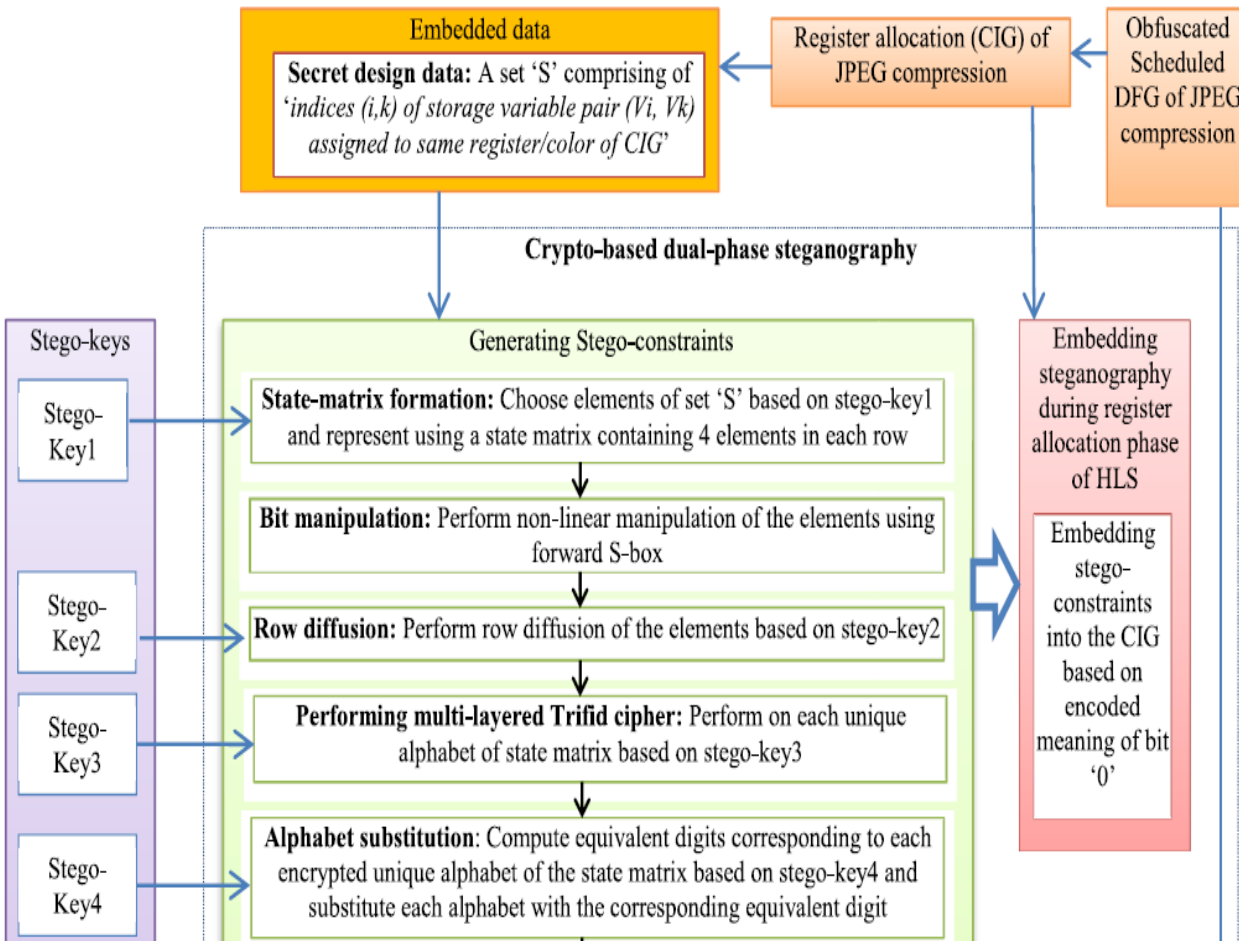


FIGURE 4. Details of 2nd line of defense: Proposed crypto-based dual-phase steganography.

Details of the Proposed Approach



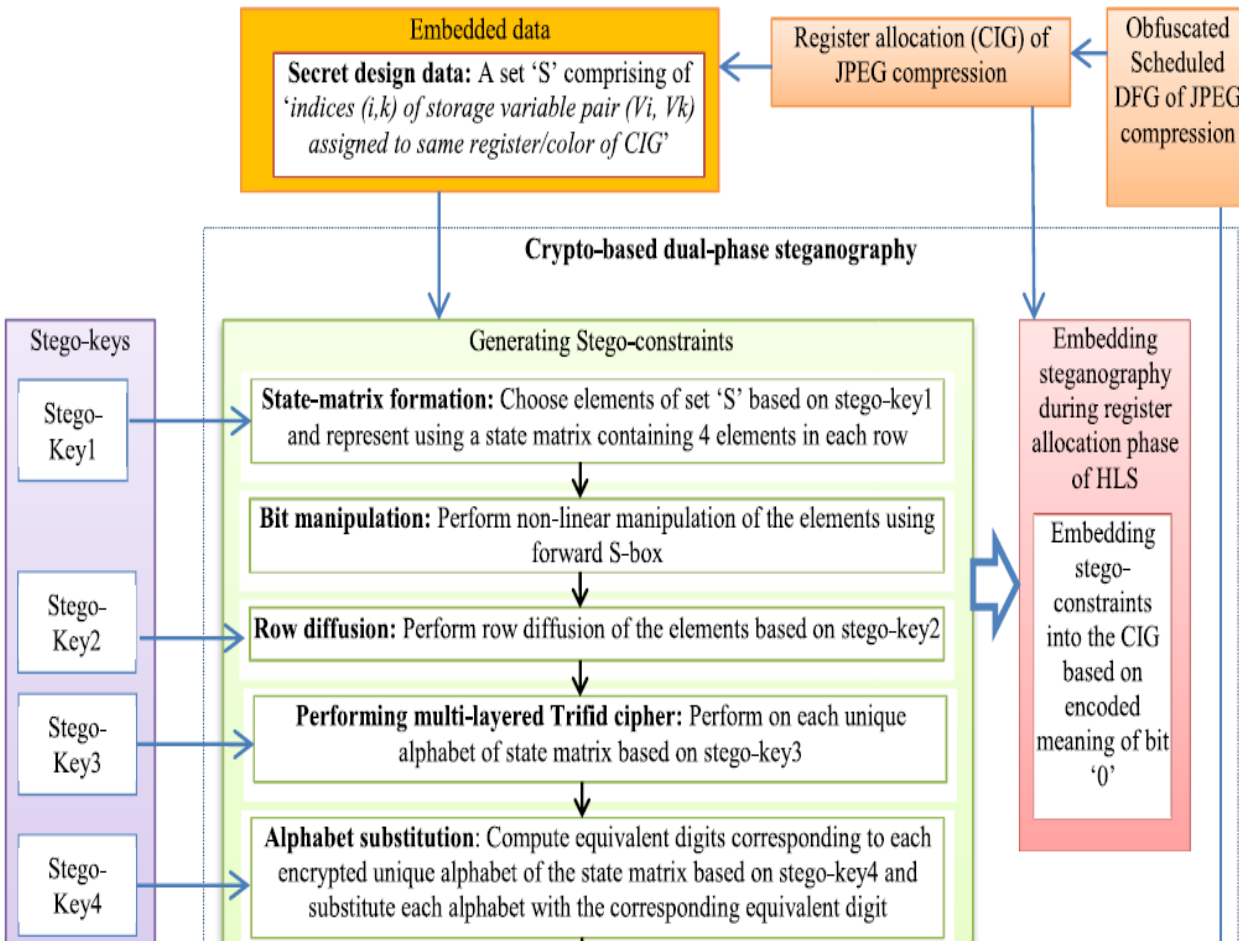
08	01	02	03
81	82	83	84
13	14	15	16
26	27	24	35
47	56	57	67

30	7C	77	7B
0C	13	EC	5F
7D	FA	59	47
F7	CC	18	96
A0	B1	5B	85

FIGURE 4. (a). Initial state matrix (b). After byte substitution

FIGURE 4. Details of 2nd line of defense: Proposed crypto-based dual-phase steganography.

Details of the Proposed Approach



30	7C	77	7B
0C	13	EC	5F
7D	FA	59	47
F7	CC	18	96
A0	B1	5B	85

→

77	7B	30	7C
5F	0C	13	EC
FA	59	47	7D
96	F7	CC	18
A0	B1	5B	85

FIGURE 4. b). After byte substitution (c). After row diffusion

The number of rows in the matrix is $R=5$, therefore the size of stego-key2 is $2 \times R = 10$ bits.

Let's assume the stego-key2 is ``01 00 10 00 11". Here, groups of two bits from left to right represent the key-bits for row-1 to row-5 respectively. Since, the key bits for rst row is ``01", Therefore circular right shift by two elements is performed for this row. Similarly, diffusion is performed for each row based on the corresponding key bits (referring Fig. 5).Post row diffusion, the modied state-matrix is shown in Fig. 9(c).

FIGURE 4. Details of 2nd line of defense: Proposed crypto-based dual-phase steganography.

Details of the Proposed Approach

Stego-key 2

Decides the number of elements (according to four modes) by which circular right shift for each row will be performed



Key-bits	Modes	Definition
00	1	Circular right shift by 1 element
01	2	Circular right shift by 2 element
01	3	Circular right shift by 3 element
11	4	Circular right shift by 4 element

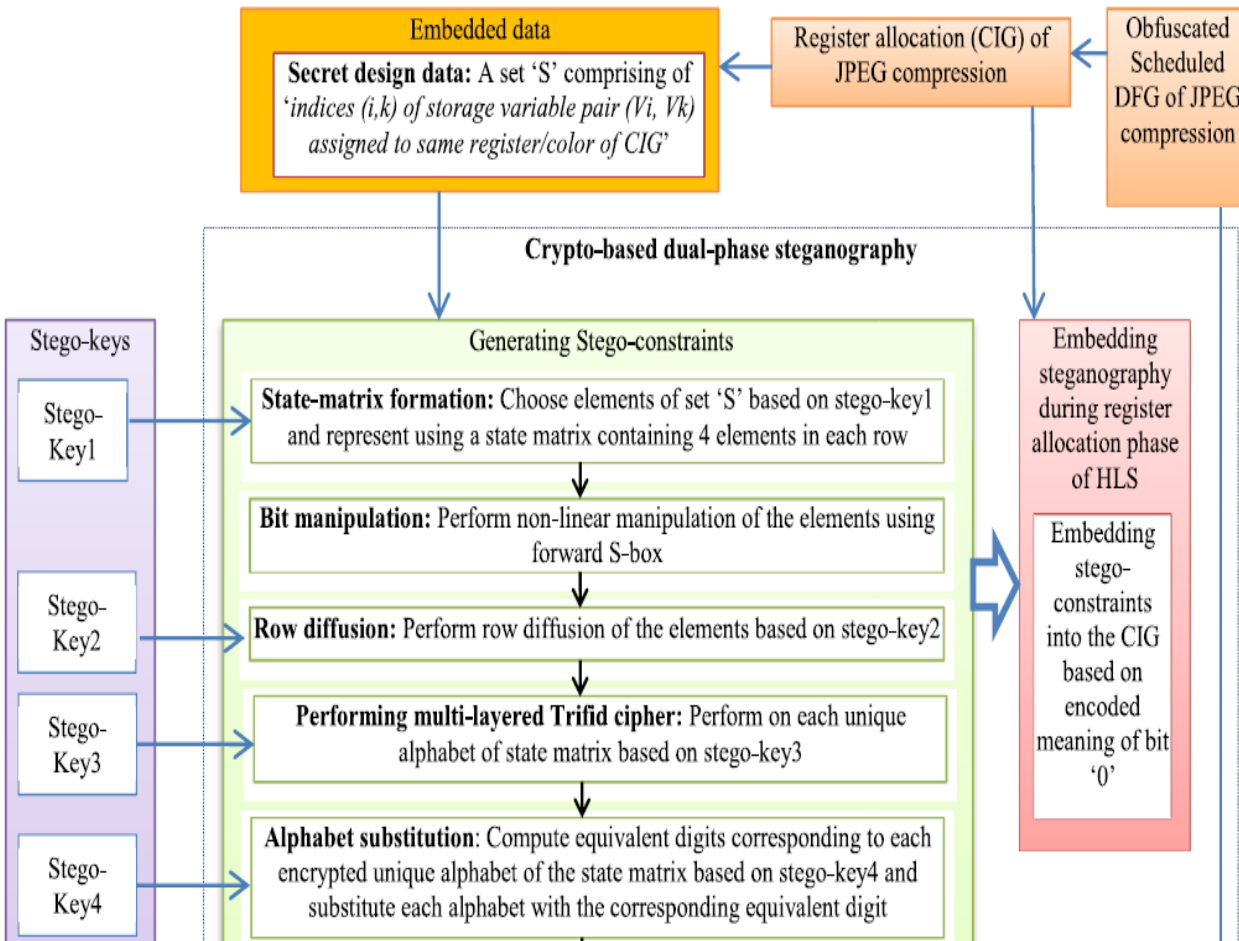
Stego-key 3

Decides the key of encryption for each unique alphabet of the state matrix (a distinct key is chosen for each unique alphabet)

Proposed algorithm uses Trifid cipher to provide following properties (i) fractionation (ii) transposition (iii) substitution.

- These properties achieve certain amount of confusion and diffusion which help in obscuring the relationship of the generated stego-constraints with the input secret design data and stego-keys.
- The stego-key3 decides the key of encryption for each unique alphabet of the state matrix.
- For each unique alphabet, the encryption key is 27 characters long. Since, there are $27!$ permutations of 27 characters, therefore $C (\log_2(27!))$ C bits are required to represent the key for an alphabet. Thus, total size of key to encipher all unique alphabets is $D (\# \text{ of unique alphabets}) d(\log_2(27!))e$. This is because, a distinct key is chosen for each unique alphabet.

Details of the Proposed Approach



77	7B	30	7C
5F	0C	13	EC
FA	59	47	7D
96	F7	CC	18
A0	B1	5B	85

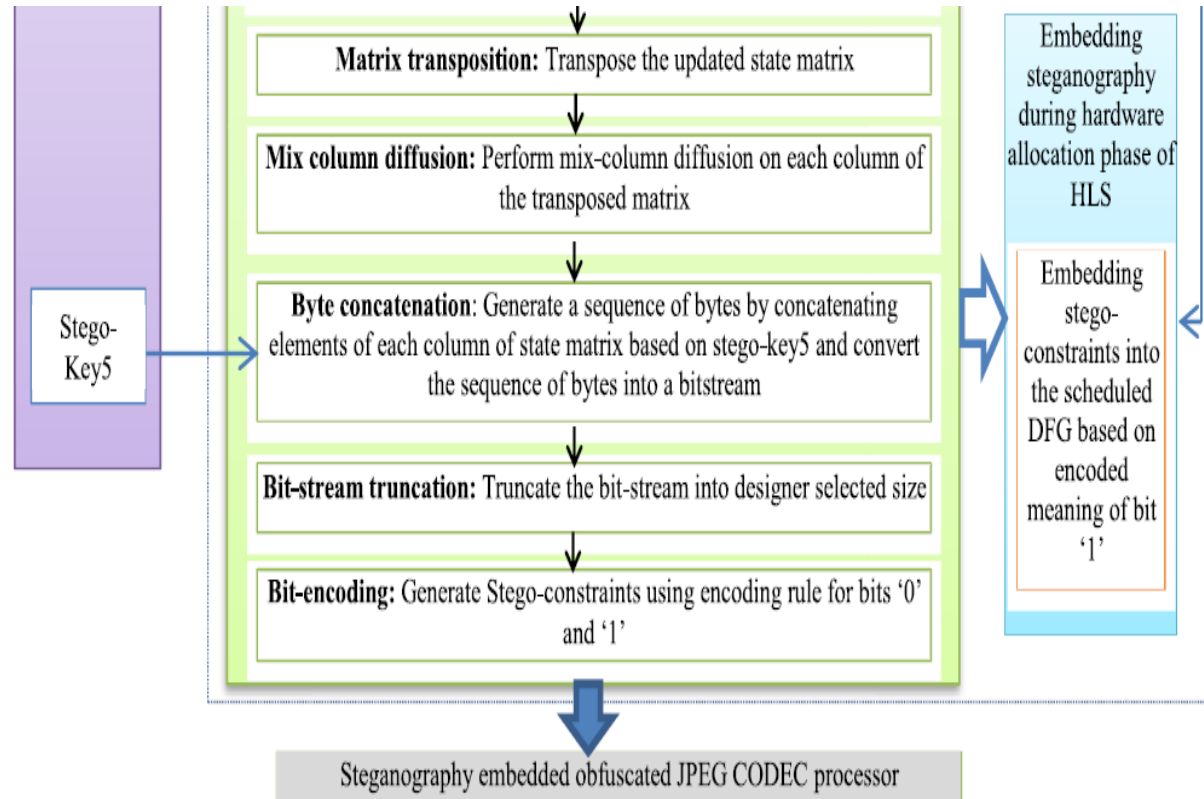
➔

77	78	30	76
51	06	13	66
14	59	47	74
96	17	66	18
40	81	58	85

FIGURE 4. (c). After row diffusion to TRIFID cipher

FIGURE 4. Details of 2nd line of defense: Proposed crypto-based dual-phase steganography.

Details of the Proposed Approach



77	78	30	76
51	06	13	66
14	59	47	74
96	17	66	18
40	81	58	85



77	51	14	96	40
78	06	59	17	81
30	13	47	66	58
76	66	74	18	85

FIGURE 4. Details of 2nd line of defense: Proposed crypto-based dual-phase steganography.

Details of the Proposed Approach

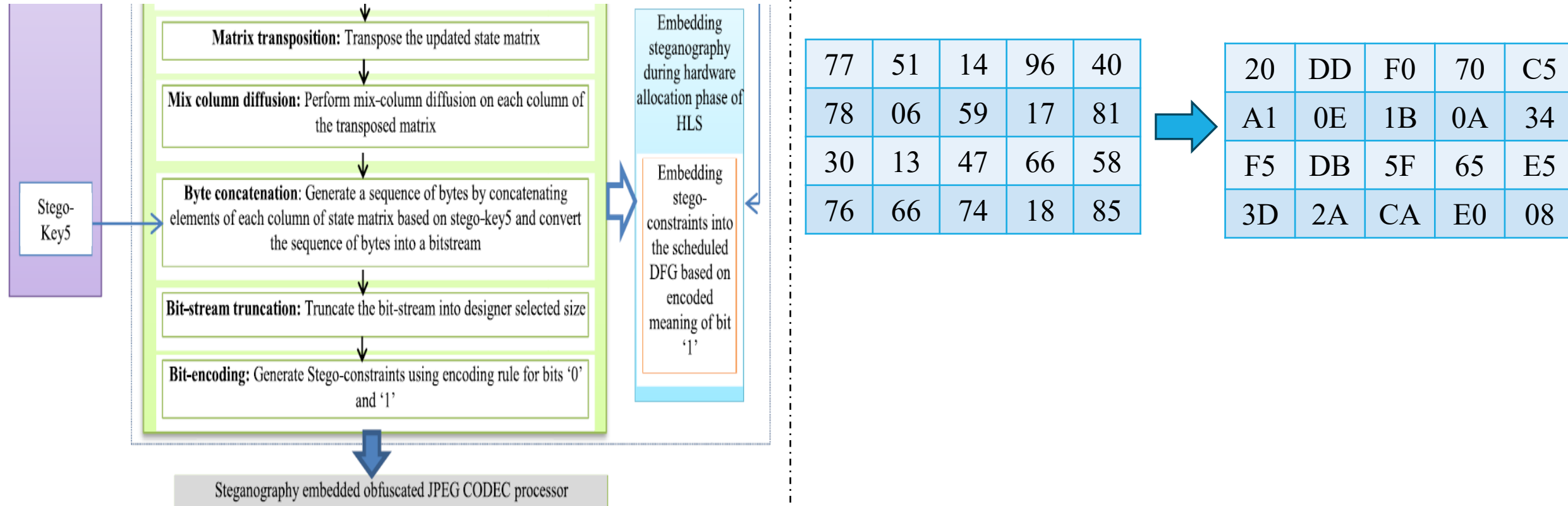
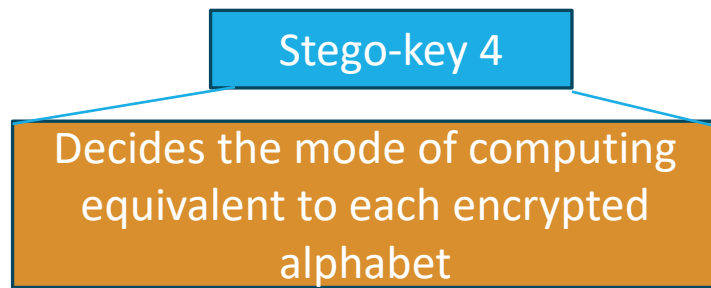
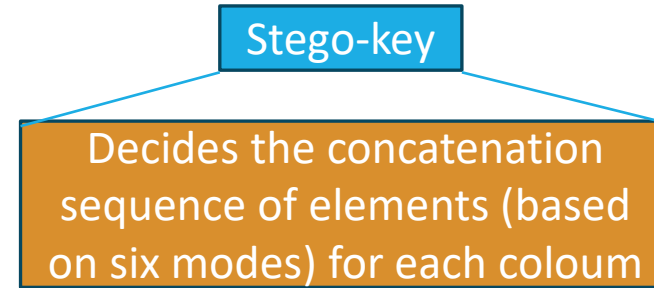


FIGURE 4. Details of 2nd line of defense: Proposed crypto-based dual-phase steganography.

Details of the Proposed Approach



Key-bits	Modes	Definition
000	1	$x*y*z$
001	2	$X+y+z$
010	3	$ x-y-z $
011	4	$ x-y+z $
100	5	$(z+y)/x$
101	6	$(z+x)*y$



Key-bits	Modes	Definition
000	1	E0E1E2E3
001	2	E0E1E3E2
010	3	E0E2E1E3
011	4	E0E2E3E1
100	5	E0E3E1E2
101	6	E0E3E2E1

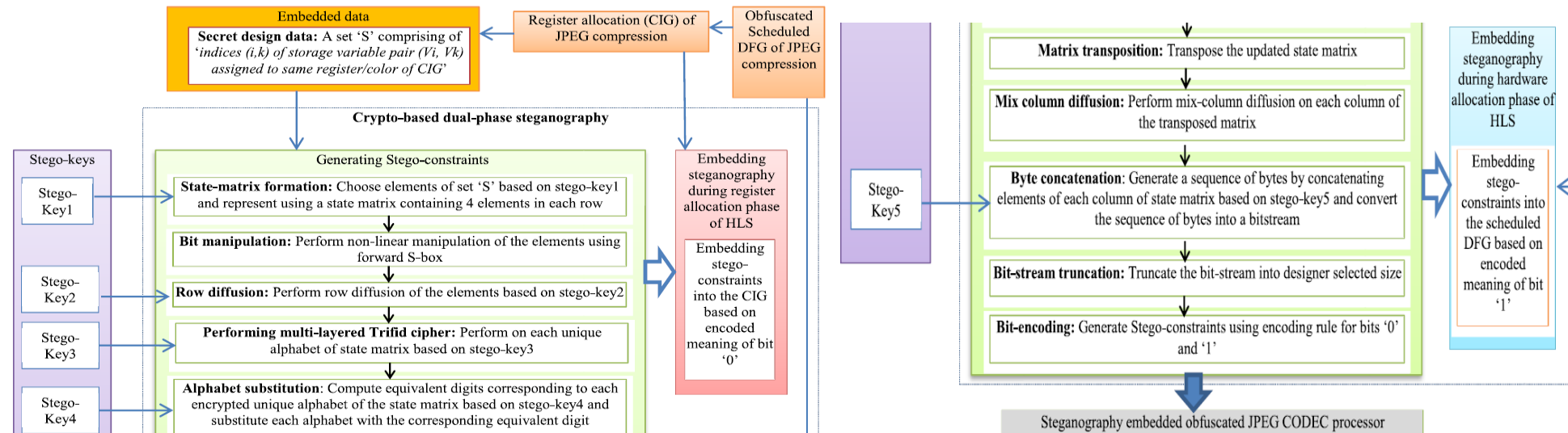
Total key size of the proposed hardware steganography =
key size of stego-key1 +
key size of stego-key2 +
key size of stego-key3 +
key size of stego-key4 +
key size of stego-key5

$$= 3\text{bits} + 2*R + (\text{\#of unique alphabets}) * [(\log_2(27!)) + (\text{\#of unique alphabets}) * [\log_2(\text{\# modes for x computing digit equivalent of unique alphabet})] + (\text{\#of columns in the transposed matrix}) * (\log_2(\text{\#of modes for concatenation}))] = 610 \text{ bits}$$

Details of the Proposed Approach

TABLE 1. Encoding of bit '0' and '1' in JPEG compression design.

Bit	Encoded meaning
0	Embed an edge between node pairs (even, even) of CIG (during register allocation of HLS) of JPEG compression
1	In the JPEG compression scheduling/allocation, odd operations are assigned to FU of vendor type 1(U1) and even operations are assigned to FU of vendor type 2(U2) (during functional unit (FU) allocation phase of HLS)



Proposed Methodology: Pseudo code of proposed double line of defense

First Line of Defense:

Inputs: DFG of JPEG compression processor, module library, resource constraints.

Output: Obfuscated scheduled DFG

Algorithm:

```
    Read DFG;
    perform_THT(){
        travers all operations;
    read data dependency of each operation;
    if operations of same type are executing sequentially,
    then execute them as parallel sub-computations while
    altering data dependency without changing functionality;
    returned obfuscated DFG;}
Read module library
Read constraints file;
architectural_synthesis()
end algorithms ;
```

Second Line of Defense:

Inputs: Obfuscated scheduled DFG, Stego-key1 to 5

Output: Steganography embedded obfuscated JPEG

Algorithm:

```
    Read obfuscated scheduled DFG ;
    create_CIG()
    extract_secret_design_data()
    state_matrix_formation()
    bit_manipulation()
    row_diffusion()
    trifid_cipher()
    alphabate_substitution()
    matrix_transposition()
    colum_diffusion()
    byte_concatenation()
    bitstream_truncation()
    bit_encoding()
    embedding_bit0()
    embedding_bit1()

end algorithms ;
```

Demonstration of Embedding Hardware Steganography on DCT core used in JPEG Compression

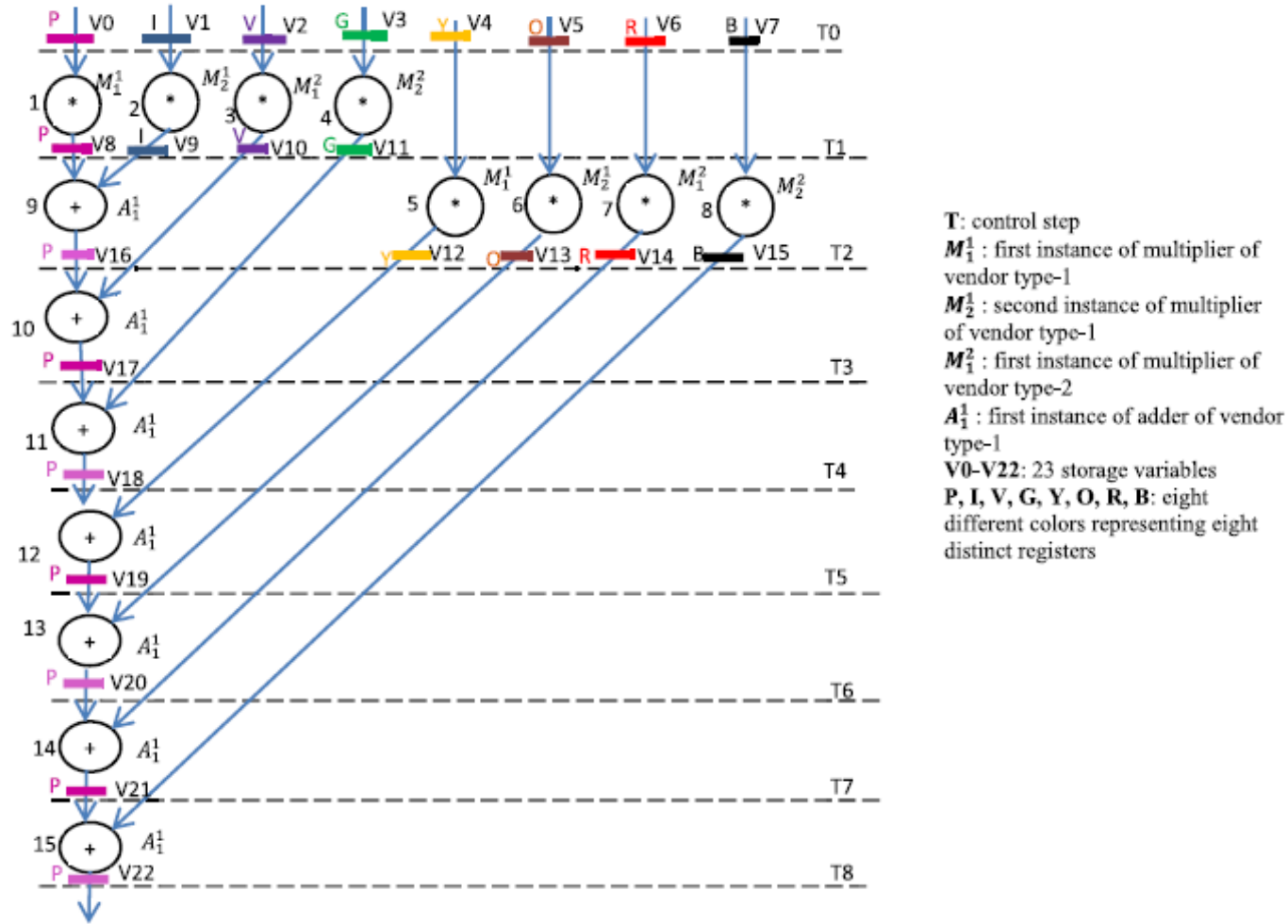


FIGURE 7. Scheduled and hardware allocated 8-point DCT using 1(+) and 4(*) before implanting steganography

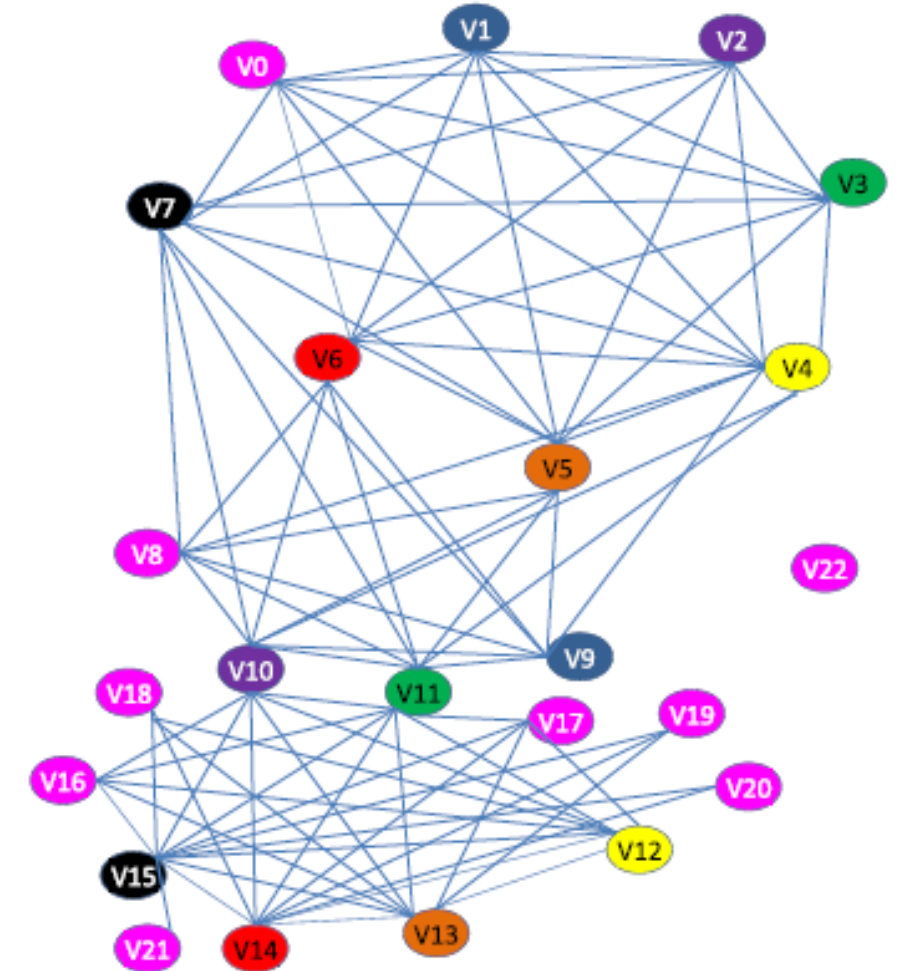


FIGURE 8. CIG of 8-point DCT before implanting hardware steganography.

Demonstration of Embedding Hardware Steganography on DCT core used in JPEG Compression

TABLE 2. Allocation of storage variables to register /colors in 8-point DCT before implanting steganography.

T	Pink	Indigo	Violet	Green	Yellow	Orange	Red	Black
0	V0	V1	V2	V3	V4	V5	V6	V7
1	V8	V9	V10	V11	V4	V5	V6	V7
2	V16	-	V10	V11	V12	V13	V14	V15
3	V17	-	-	V11	V12	V13	V14	V15
4	V18	-	-	-	V12	V13	V14	V15
5	V19	-	-	-	-	V13	V14	V15
6	V20	-	-	-	-	-	V14	V15
7	V21	-	-	-	-	-	-	V15
8	V22	-	-	-	-	-	-	-

A secret design data extracted from CIG is represented by a set 'S' as follows:

$S = \{(0,8), (0,16), (0,17), (0,18), (0,19), (0,20), (0,21), (0,22), (8,16), (8,17), (8,18), (8,19), (8,20), (8,21), (8,22), (16,17), (16,18), (16,19), (16,20), (16,21), (16,22), (17,18), (17,19), (17,20), (17,21), (17,22), (18,19), (18,20), (18,21), (18,22), (19,20), (19,21), (19,22), (20,21), (20,22), (21,22), (1,9), (2,10), (3,11), (4,12), (5,13), (6,14), (7,15)\}$

Demonstration of Embedding Hardware Steganography on DCT core used in JPEG Compression

(a)				(b)				(c)			
08	01	02	03	30	7C	77	7B	77	7B	30	7C
81	82	83	84	0C	13	EC	5F	5F	0C	13	EC
13	14	15	16	7D	FA	59	47	FA	59	47	7D
26	27	24	35	F7	CC	18	96	96	F7	CC	18
47	56	57	67	A0	B1	5B	85	A0	B1	5B	85

FIGURE 9. (a). Initial state matrix

(b). After byte substitution

(c). After row diffusion.

(a)				(b)					(c)				
77	78	30	76	77	51	14	96	40	20	DD	F0	70	C5
51	06	13	66	78	06	59	17	81	A1	0E	1B	0A	34
14	59	47	74	30	13	47	66	58	F5	DB	5F	65	E5
96	17	66	18	76	66	74	18	85	3D	2A	CA	E0	08
40	81	58	85										

FIGURE 10. (a). After TRIFID cipher.

(b). After transposition.

(c). After mix-column diffusion.

Demonstration of Embedding Hardware Steganography on DCT core used in JPEG Compression

TABLE 3. Digits equivalent of alphabets computed based on stego-key4.

Alphabets	A	B	C	D	E	F
Corresponding state (output of TRIFID cipher)	211	323	321	233	313	322
Key bits (Stego-Key4)	001	001	000	010	101	010
Computed digit	4	8	6	4	6	1

$$\begin{pmatrix} E0 \\ E1 \\ E2 \\ E3 \end{pmatrix} = \begin{pmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{pmatrix} * \begin{pmatrix} 77 \\ 78 \\ 30 \\ 76 \end{pmatrix} = \begin{pmatrix} 20 \\ A1 \\ F5 \\ 3D \end{pmatrix}$$

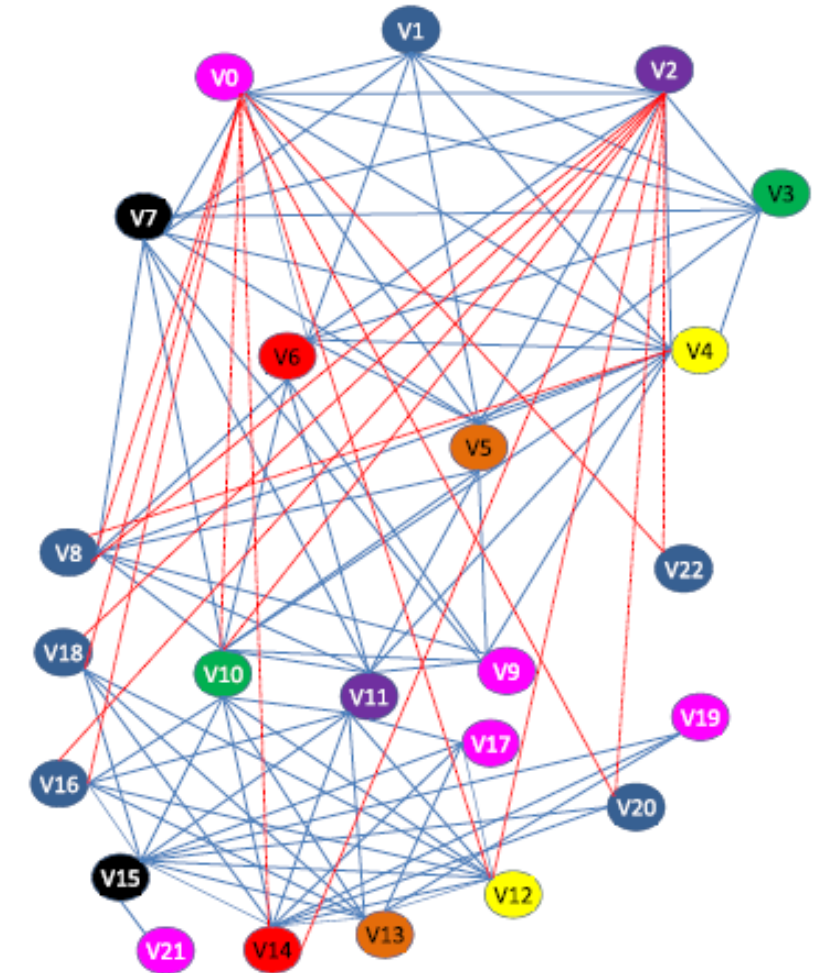


FIGURE 11. The CIG of 8-point DCT after implanting steganography during register allocation phase Note: Dotted red lines show the stego-constraints implanted (corresponding to 0s).

Demonstration of Embedding Hardware Steganography on DCT core used in JPEG Compression

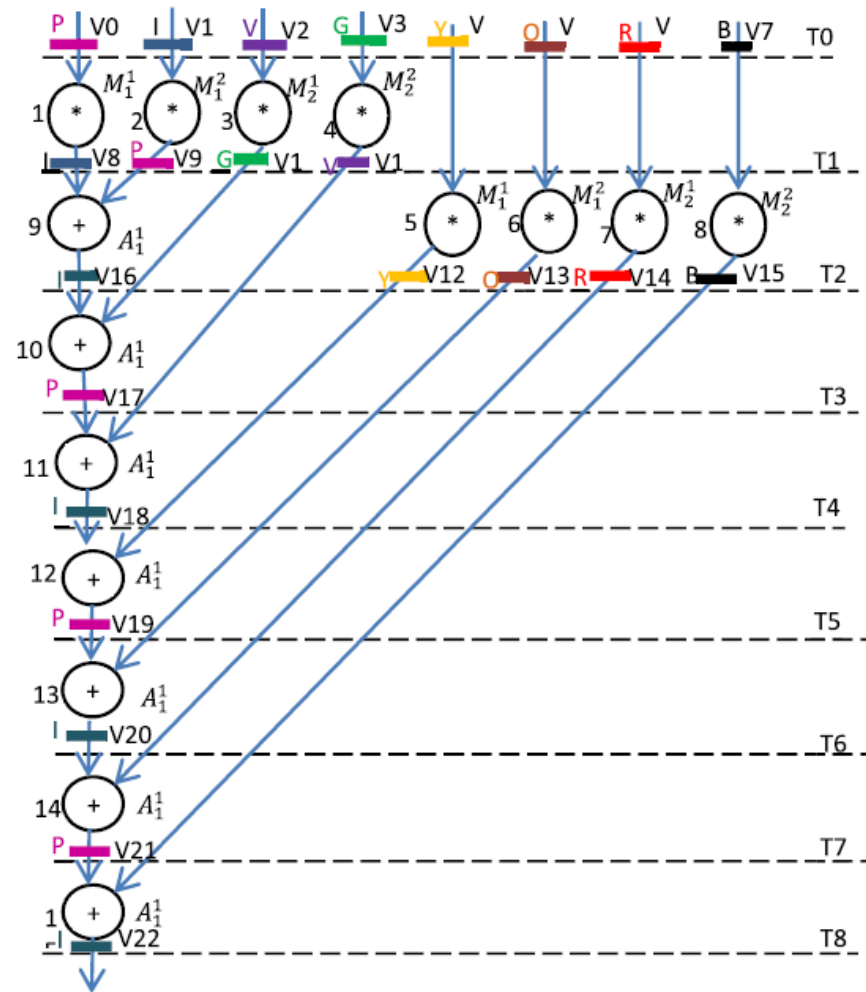


FIGURE 12. Scheduled and hardware allocated 8-point DCT using 1(+) and 4(*) after implanting steganography during both phase-1 and phase-2.

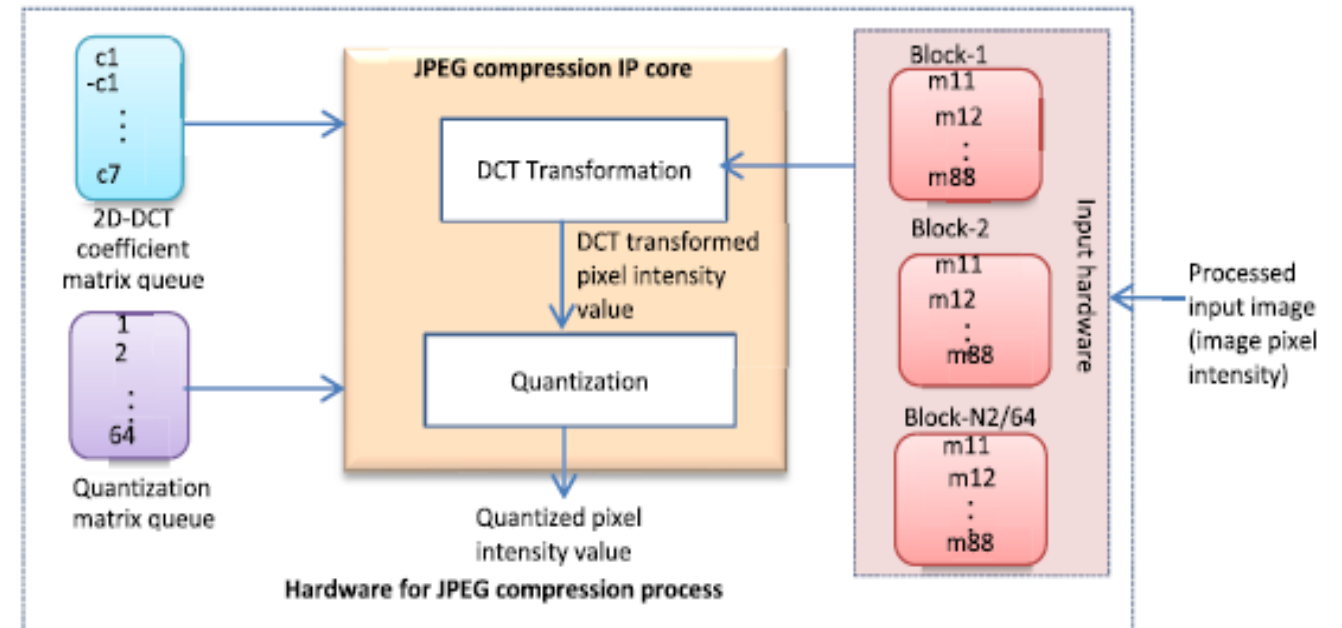


FIGURE 13. Hardware of JPEG compression process.

Designing JPEG CODEC with Embedded Hardware Steganography

TABLE 4. Register/color allocation of storage variables of 8-point DCT after implanting steganography.

C4	C4	C4	C4	C4	C4	C4	C4
C1	C3	C5	C7	-C7	-C5	-C3	-C1
C2	C6	-C6	-C2	-C2	-C6	-C6	C2
C3	-C7	-C1	-C5	C5	C1	C7	-C3
C4	-C4	-C4	C4	C4	-C4	-C4	C4
C5	-C1	C7	C3	-C3	-C7	C1	-C5
C6	-C2	C2	-C6	-C6	C2	-C2	C6
C7	-C5	C3	-C1	C1	-C3	C5	-C7

T	Pink	Indigo	Violet	Green	Yellow	Orange	Red	Black
0	V0	V1	V2	V3	V4	V5	V6	V7
1	V9	V8	V11	V10	V4	V5	V6	V7
2	--	V16	V11	V10	V12	V13	V14	V15
3	V17	--	V11	--	V12	V13	V14	V15
4	--	V18	--	--	V12	V13	V14	V15
5	V19	--	--	--	--	V13	V14	V15
6	--	-V20	--	--	--	--	V14	V15
7	V21	--	--	--	--	--	--	V15
8	--	V22	--	--	--	--	--	--

FIGURE 14. 2D- DCT coefficient matrix.

Designing JPEG CODEC with Embedded Hardware Steganography

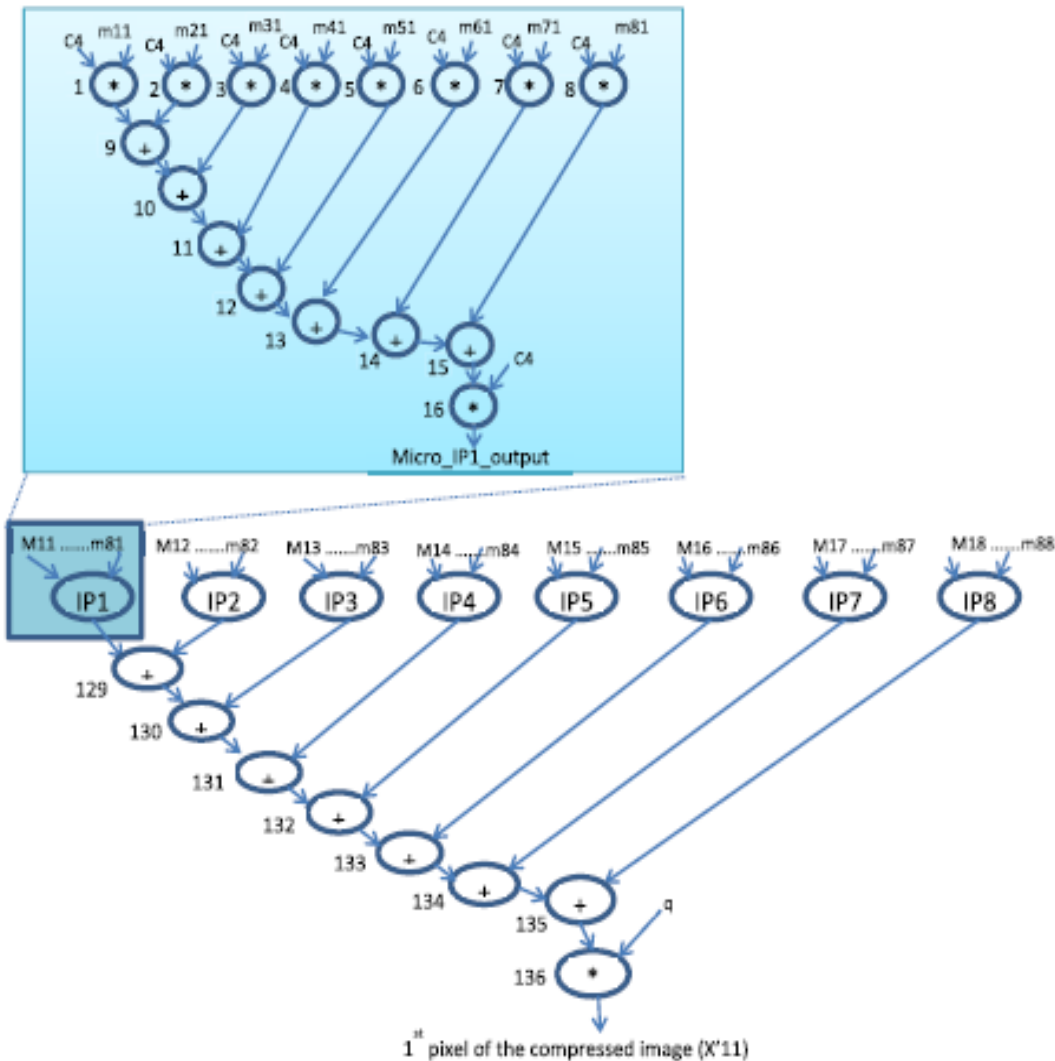


FIGURE 15. DFG of un-obfuscated JPEG compression IP core as macro IP comprising 8 micro-IPs underneath.

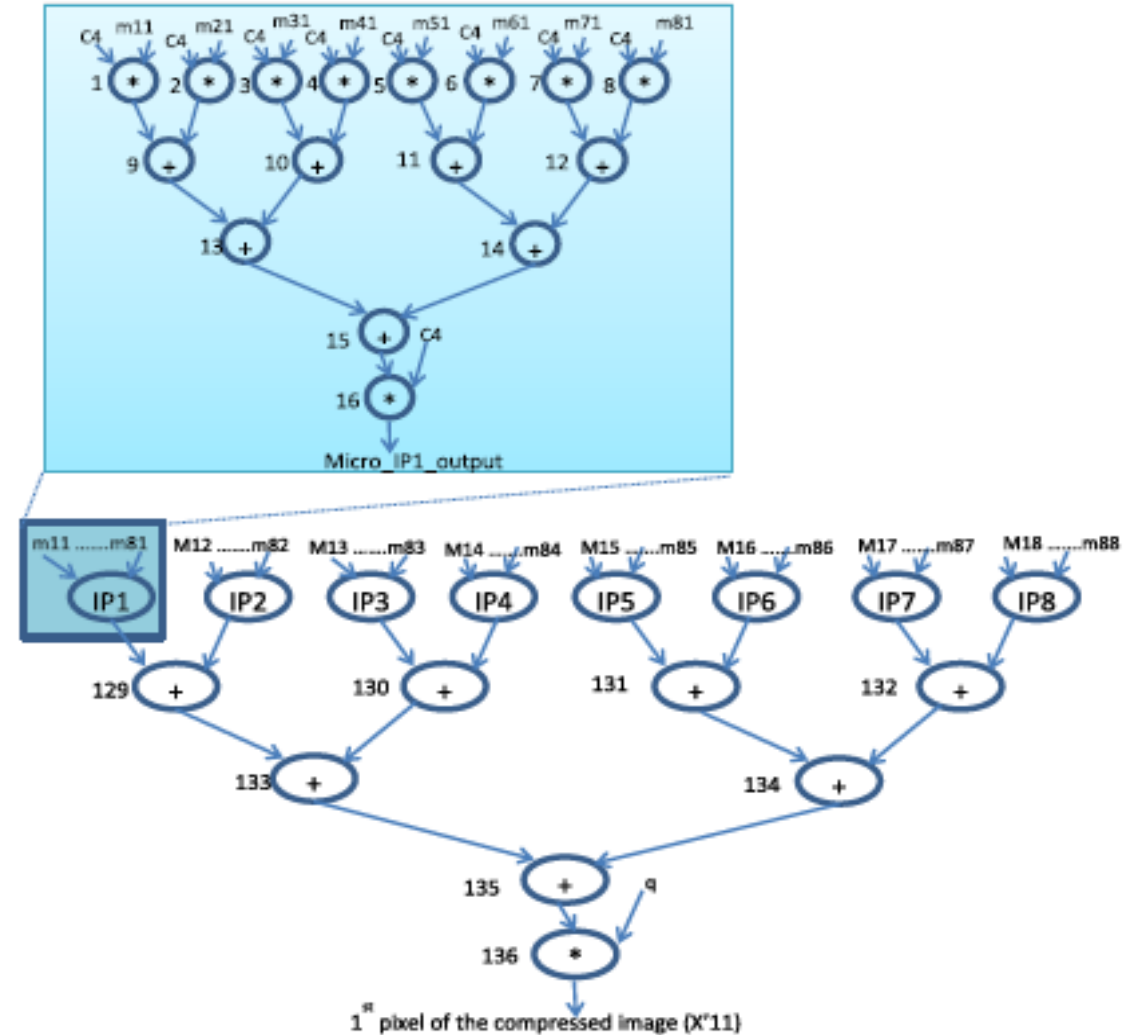


FIGURE 16. Structurally obfuscated DFG of JPEG CODEC IP core.

Designing JPEG CODEC with Embedded Hardware Steganography

TABLE 6. Scheduling of JPEG compression hardware after implanting steganography.

T	Operations assign to M_1^1	Operations assign to M_2^1	Operations assign to M_1^2	Operations assign to A_1^1	Operations assign to A_2^1	Operations assign to A_1^2
1	1	3	2	--	--	--
2	5	6	4	9	--	--
3	7	17	8	11	--	10
4	19	20	18	13	--	12
5	21	23	22	25	26	14
6	33	34	24	15	27	29
7	35	37	36	41	--	28
8	39	40	38	42	--	30
9	49	51	50	43	45	44
10	53	54	52	31	57	46
11	55	65	56	59	47	58
12	67	68	66	61	--	60
13	69	71	70	73	74	62
14	81	82	72	63	75	77
15	83	85	84	89	--	76
16	87	88	86	90	--	78
17	97	99	98	91	93	92
18	101	102	100	79	105	94
19	103	113	104	95	107	106
20	115	116	114	--	109	108
21	117	119	118	121	122	110
22	32	120	16	111	123	125
23	64	80	48	129	--	124
24	112	--	96	130	--	126
25	--	--	--	127	131	133
26	--	--	128	--	--	--
27	--	--	--	--	--	132
28	--	--	--	--	--	134
29	--	--	--	135	--	--
30	--	136	--	--	--	--

Designing JPEG CODEC with Embedded Hardware Steganography

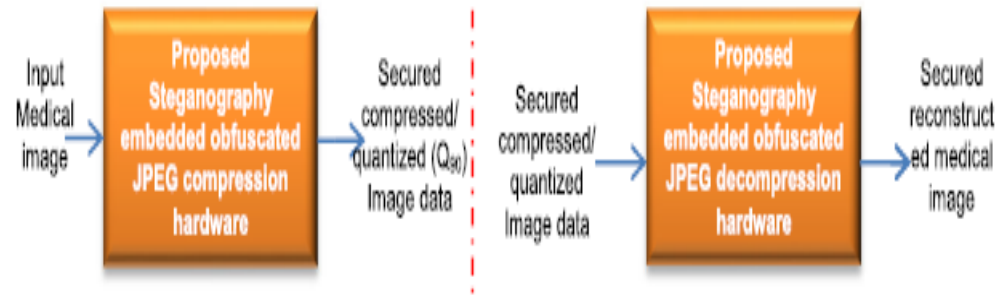


FIGURE 18. End to end JPEG CODEC process.

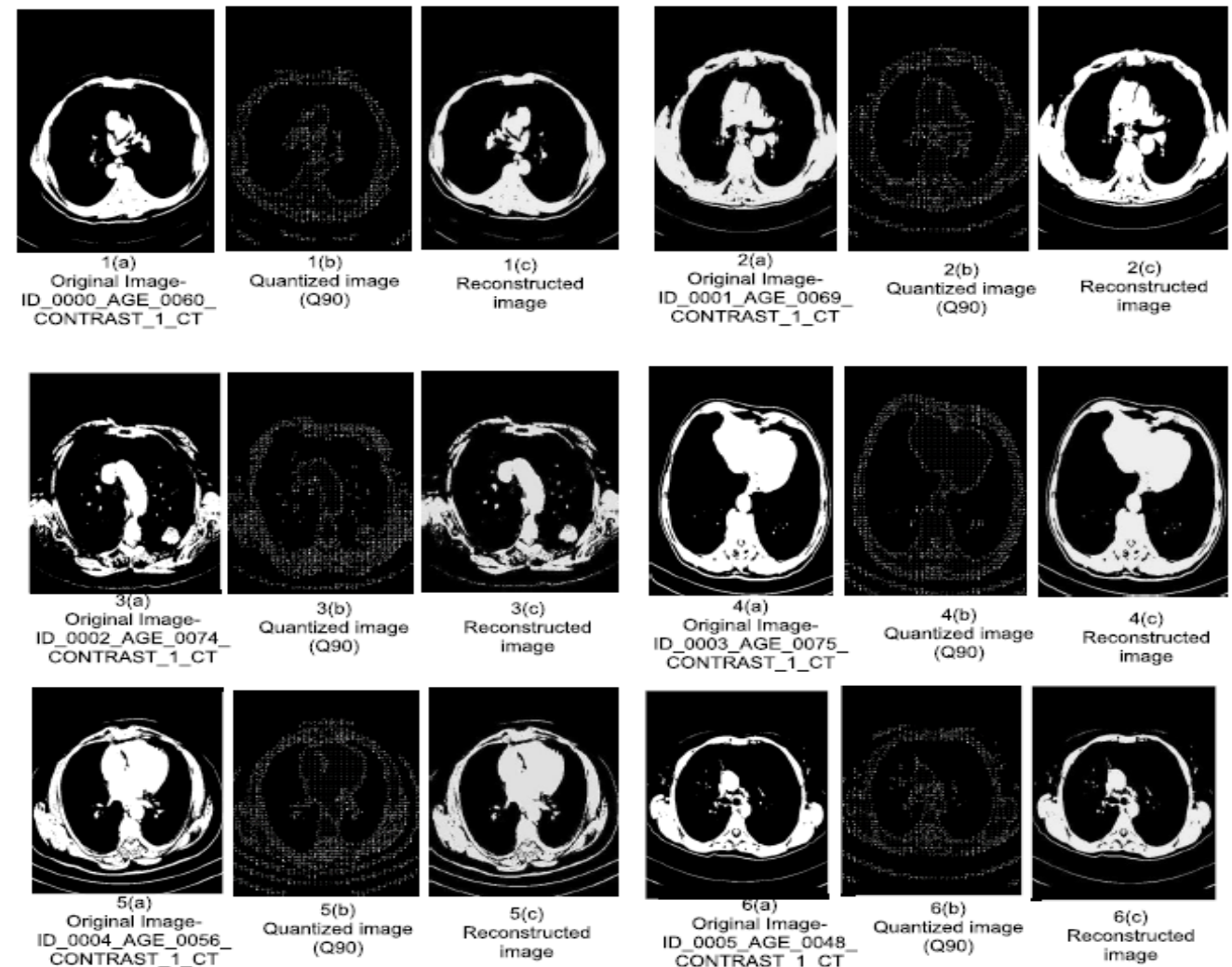


FIGURE 19. Original, quantized (at quality level Q90) and reconstructed CT scan images using JPEG CODEC hardware.

Experimental Results

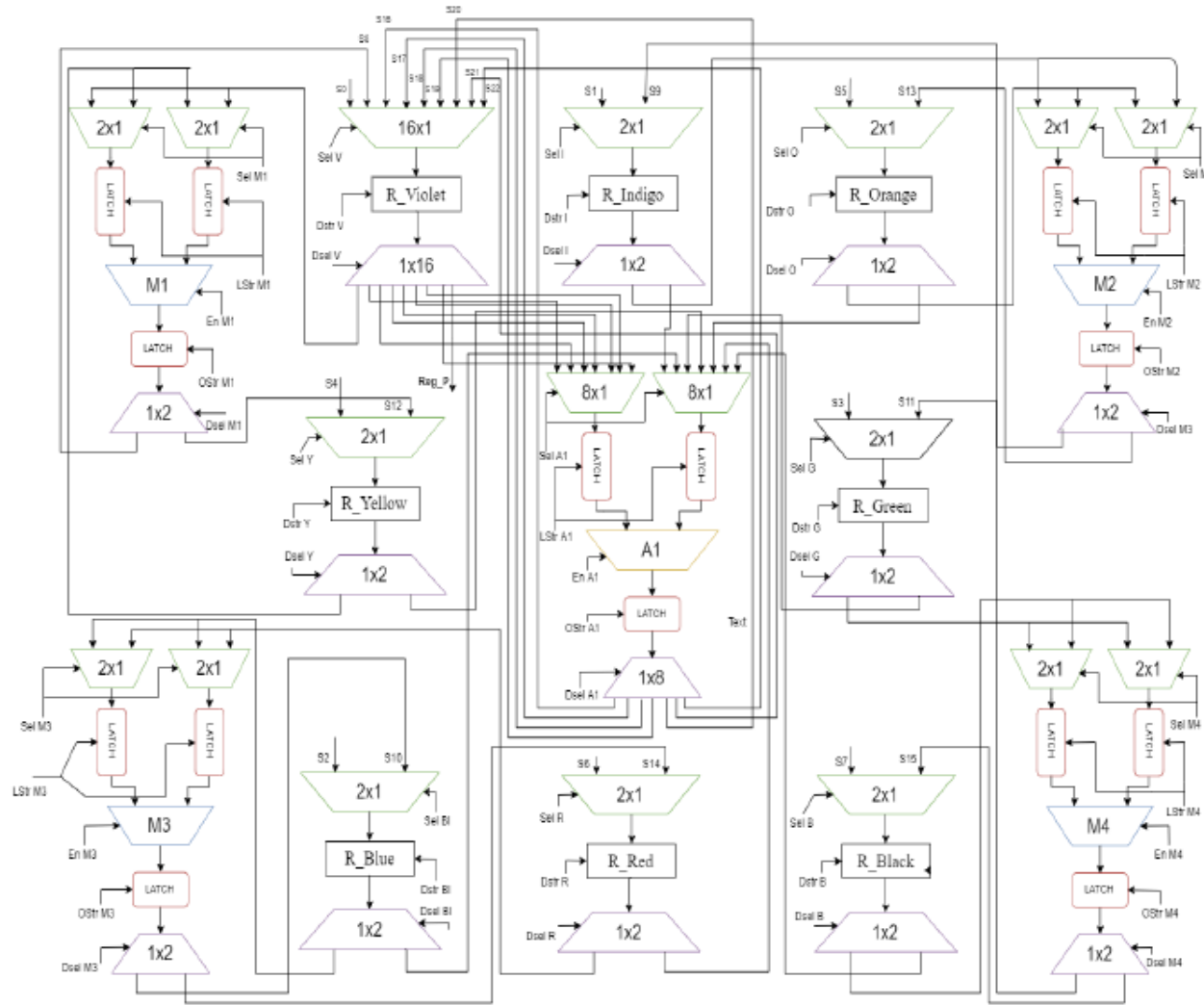


FIGURE 20. Un-obfuscated JPEG DCT core.

TABLE 7. Results of compressed CT scan images [19] using proposed secured JPEG CODEC.

Original Image Name	PSNR	MSE
ID_0000_AGE_0060_CONTRAST_1_CT	22.5965	1.2375
ID_0001_AGE_0069_CONTRAST_1_CT	21.5139	1.5878
ID_0002_AGE_0074_CONTRAST_1_CT	20.0345	2.2322
ID_0003_AGE_0075_CONTRAST_1_CT	21.1257	1.7362
ID_0004_AGE_0056_CONTRAST_1_CT	19.1805	2.7173
ID_0005_AGE_0048_CONTRAST_1_CT	22.1125	1.3834

Experimental Results

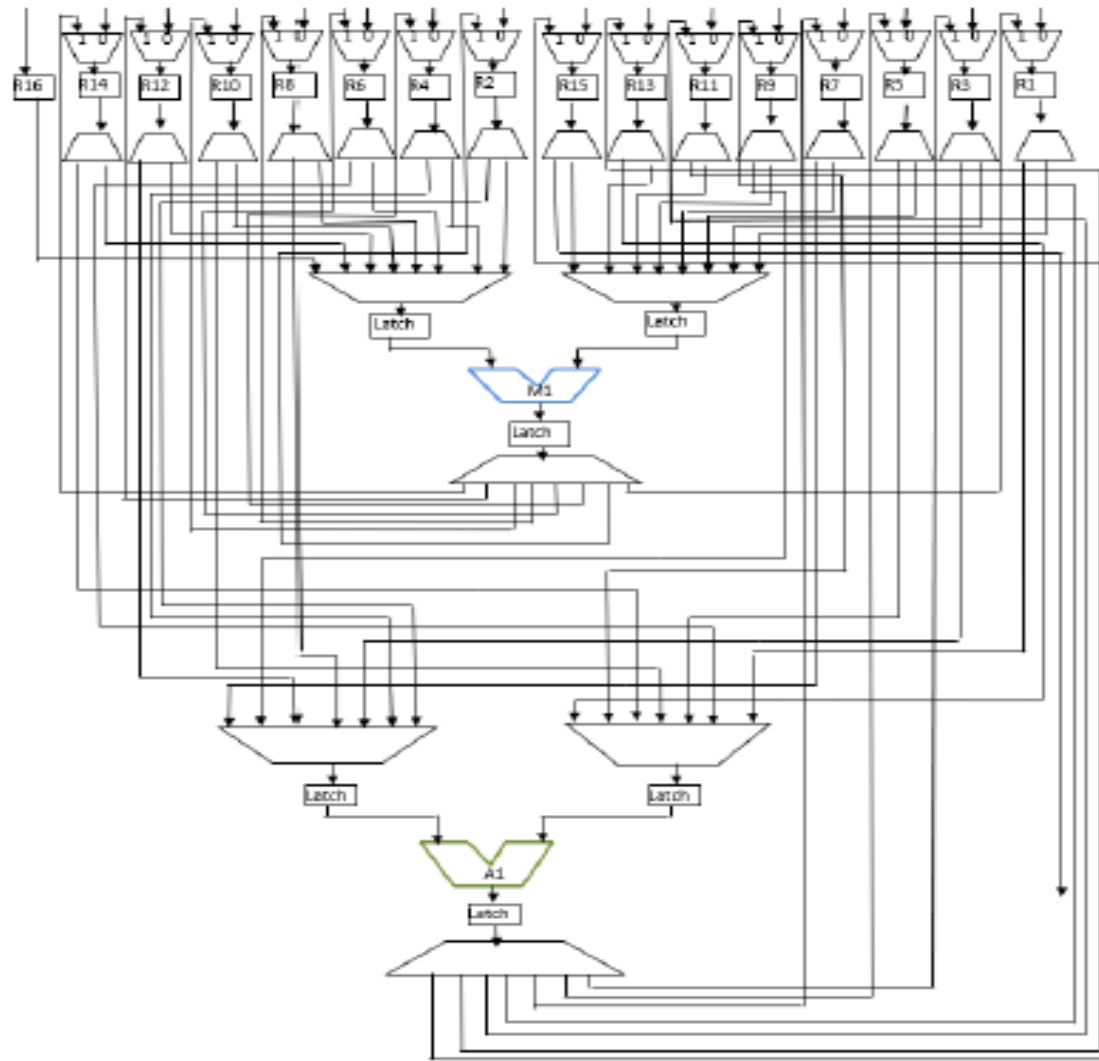


FIGURE 21. Obfuscated JPEG DCT core.

Experimental Results

TABLE 8. Comparison of JPEG hardware pre and post structural obfuscation.

	Resource Configuration	Structural changes due to proposed obfuscation
Non-obfuscated JPEG hardware	4+, 8*, 12(8:1 mux), 12(16:1 mux), 6(1:8 demux), 6(1:16 demux)	10064 gates
Structurally obfuscated JPEG hardware	3+, 3*, 10(32:1 mux), 2(16:1 mux), 5(1:32 demux), 1(1:16 demux)	

TABLE 9. Effective number of 0's and 1's for different size of stego- constraints

Design solution (resource constraint)	Total # of constraint =100		Total # of constraint =200		Total # of constraint =300		Total # of constraint =400	
	Effective # of 0's embedded	Effective # of 1's embedded	Effective # of 0's embedded	Effective # of 1's embedded	Effective # of 0's embedded	Effective # of 1's embedded	Effective # of 0's embedded	Effective # of 1's embedded
3+, 3*	42	49	89	93	139	111	197	111
3+, 5*	48	48	97	94	148	122	203	122
5+, 5*	49	49	98	93	153	124	208	124
7+, 9*	40	59	87	108	135	131	186	131
9+, 9*	44	55	93	104	142	132	188	132
11+, 11*	53	46	109	89	160	131	217	131

Experimental Results

TABLE 10. Security analysis (in terms of probability of coincidence) of proposed approach on varying size of stego-constraints for different design solutions.

Design solution	# of constraint =100		# of constraint =200		# of constraint =300		# of constraint =400	
	Pc ¹	Pc ²	Pc ¹	Pc ²	Pc ¹	Pc ²	Pc ¹	Pc ²
3+, 3*	5.21 e-1	1.6245e-3	2.5161e-1	4.4e-6	1.1589e-1	2.4e-7	4.715e-2	9.89e-8
3+, 5*	4.75e-1	1.732e-2	2.222e-1	3.39e-4	1.008e-1	2.228e-5	4.296e-1	9.497e-6
5+, 5*	4.68e-1	6.329e-2	2.188e-1	4.913e-3	9.328e-2	5.907e-4	3.976e-2	2.518e-4
7+, 9*	5.38e-1	2.092e-1	2.595e-1	4.61e-2	1.233e-1	1.515e-2	5.59e-2	6.87e-3
9+, 9*	5.06e-1	2.552e-1	2.364e-1	6.496e-2	1.106e-1	2.146e-2	5.42e-2	1.051e-2
11+, 11*	4.40e-1	3.001e-1	1.845e-1	8.816e-2	8.368e-2	2.821e-2	3.458e-2	1.166e-2

TABLE 11. Design cost analysis of proposed approach on varying size of stego-constraints for different design solutions.

Design solution	Pre-stegano-graphy cost	# of constraint= 100		# of constraint= 200		# of constraint= 300		# of constraint= 400	
		Cost (Phase-1)	Cost (Phase-1 & Phase-2)	Cost (Phase-1)	Cost (Phase-1 & Phase-2)	Cost (Phase-1)	Cost (Phase-1 & Phase-2)	Cost (Phase-1)	Cost (Phase-1 & Phase-2)
3+, 3*	0.2167	0.2167	0.2167	0.2167	0.2169	0.2167	0.2173	0.2167	0.2173
3+, 5*	0.1917	0.1917	0.1920	0.1917	0.1924	0.1917	0.1929	0.1917	0.1929
5+, 5*	0.1713	0.1713	0.1713	0.1713	0.1713	0.1713	0.1719	0.1713	0.1719
7+, 9*	0.1718	0.1718	0.1720	0.1718	0.1725	0.1718	0.1729	0.1718	0.1729
9+, 9*	0.1752	0.1752	0.1754	0.1752	0.1757	0.1752	0.1763	0.1752	0.1763
11+, 11*	0.1785	0.1785	0.1785	0.1785	0.1789	0.1785	0.1794	0.1785	0.1794

TABLE 12. Comparison of design cost and security of JPEG DCT core using proposed approach and [21].

Comparison metric	Proposed	[21]
Design cost	0.45	0.45
Security in terms of key size (in bits)	610	0

References

- [1] D. A. Koff and H. Shulman, "An overview of digital compression of medical images: Can we use lossy image compression in radiology?" Can. Assoc. Radiol. J., vol. 57, no. 4, pp. 211217, 2006.
- [2] S. Gokturk, C. Tomasi, B. Girod, and C. Beaulieu, "Medical image compression based on region of interest, with application to colon CT images," in Proc. Conf. 23rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc., vol. 3, Aug. 2005, pp. 24532456.
- [3] S. B. Gokturk. Region of Interest Based Medical Image Compression. Accessed: Aug. 2019. [Online]. Available: <http://ai.stanford.edu/~gokturkb/Compression/FinalReport.htm>
- [4] Y.-Y. Chen and S.-C. Ti, "Embedded medical image compression using DCT based subband decomposition and modified SPIHT data organization," in Proc. 4th IEEE Symp. Bioinf. Bioeng., Oct. 2004, pp. 167174.
- [5] Y.-Y. Chen, "Medical image compression using DCT-based subband decomposition and modified SPIHT data organization," Int. J. Med. Inform., vol. 76, no. 10, pp. 717725, Oct. 2007.
- [6] R. Agarwal, C. S. Salimath, and K. Alam, "Multiple image compression in medical imaging techniques using wavelets for speedy transmission and optimal storage," Biomed. Pharmacol. J., vol. 12, no. 1, pp. 183198, Mar. 2019.
- [7] X. Zhang and M. Tehranipoor, "Case study: Detecting hardware trojans in third-party digital IP cores," in Proc. IEEE Int. Symp. Hardw.-Oriented Secur. Trust, San Diego CA, USA, Jun. 2011, pp. 6770.
- [8] Maxim. Accessed: May. [Online]. Available: <https://www.maximintegrated.com/en/app-notes/index.mvp/id/545>
- [9] SMT Corp. Counterfeit Detection. Accessed: May 2019. [Online]. Available: <https://www.smtcorp.com/counterfeit-detection>
- [10] A. Sengupta, D. Roy, and S. P. Mohanty, "Triplephase watermarking for reusable ip core protection during architecture synthesis," IEEE Trans. Comput.-Aided Design Integr. Circuits Syst., vol. 37, no. 4, pp. 742755, Apr. 2018.
- [11] A. Sengupta, D. Roy, S. P. Mohanty, and P. Corcoran, "DSP design protection in CE through algorithmic transformation based structural obfuscation," IEEE Trans. Consum. Electron., vol. 63, no. 4, pp. 467476, Nov. 2017.
- [12] Y. Lao and K. K. Parhi, "Obfuscating DSP circuits via highlevel transformations," IEEE Trans. Very Large Scale Integr. (VLSI) Syst., vol. 23, no. 5, pp. 819830, May 2015.
- [13] A. Sengupta, D. Roy, S. P. Mohanty, and P. Corcoran, "Low-cost obfuscated JPEG CODEC IP core for secure CE hardware," IEEE Trans. Consum. Electron., vol. 64, no. 3, pp. 365374, Aug. 2018.
- [14] A. Sengupta, R. Sedaghat, and Z. Zeng, "A high level synthesis design flow with a novel approach for efficient design space exploration in case of multi-parametric optimization objective," Microelectron. Rel., vol. 50, no. 3, pp. 424437, Mar. 2010.
- [15] M. C. McFarland, A. C. Parker, and R. Camposano, "Tutorial on high level synthesis," in Proc. 25th ACM/IEEE Design Autom. Conf. (DAC), Hoboken, NJ, USA, Jun. 1988, pp. 330336.
- [16] A. Sengupta and S. Bhadauria, "Exploring low-cost optimal watermark for reusable IP cores during high level synthesis," IEEE Access, vol. 4, pp. 21982215, 2016.
- [17] F. Koushanfar, I. Hong, and M. Potkonjak, "Behavioral synthesis techniques for intellectual property protection," ACM Trans. Des. Autom. Electron. Syst., vol. 10, no. 3, pp. 523545, Jul. 2005.
- [18] NanGate 15 nm Open Cell Library. Accessed: Jun. 2019. [Online]. Available: <http://www.nangate.com/?pageid=2328>
- [19] Kagal. CT Medical Images. Accessed: Jul. 2019. [Online]. Available: <https://www.kaggle.com/kmader/siim-medical-images/home>
- [20] A. Sengupta and M. Rathor, "Protecting DSP kernels using robust hologrambased obfuscation," IEEE Trans. Consum. Electron., vol. 65, no. 1, pp. 99108, Feb. 2019.
- [21] A. Sengupta and M. Rathor, "IP core steganography for protecting DSP kernels used in CE systems," IEEE Trans. Consum. Electron., vol. 65, no. 4, pp. 506515, Nov. 2019.
- [22] B. Le Gal and L. Bossuet, "Automatic low-cost IP watermarking technique based on output mark insertions," Des. Autom. Embedded Syst., vol. 16, no. 2, pp. 7192, Jun. 2012.
- [23] A. Sengupta and D. Roy, "Anti-piracy aware IP chipset design for CE devices: Robust watermarking approach," IEEE Consum. Electron. Mag., vol. 6, no. 2, pp. 24118, 2017.

THANK YOU